Accelerating Critic Learning via Lyapunov-Structured Value Functions for Reinforcement Learning

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Anthropic, Gemini & OpenAI servers. Planet Earth.

Abstract

Learning accurate value functions from scratch is a key challenge contributing to the sample inefficiency of deep reinforcement learning in continuous control. To address this, we investigate incorporating control-theoretic priors by structuring the critic's value function as the sum of a known analytic Lyapunov function and a learned neural network residual. We evaluated this approach using the Proximal Policy Optimization (PPO) algorithm on the Gymnasium Pendulum-v1 stabilization task, comparing a standard agent against one with the Lyapunov-structured critic. Our results show that the structured critic converged substantially faster, achieving an 87% lower overall training loss and an 8-fold reduction in loss during early training compared to the baseline. Furthermore, the resulting value function was 86% closer to the analytic Lyapunov function. However, these significant improvements in value function approximation did not translate into superior policy performance or sample efficiency within the 100,000-step training horizon, as neither agent learned a stable policy. These findings suggest that while Lyapunov structural priors can dramatically accelerate value function convergence, the realization of corresponding policy improvements in on-policy algorithms may require a more extensive training budget.

1 Introduction

Deep reinforcement learning has emerged as a powerful paradigm for solving complex sequential decision-making problems. However, its application to continuous control tasks, particularly in robotics and physical systems, is often hampered by substantial sample inefficiency. A primary contributor to this challenge is the difficulty of learning an accurate value function, which estimates the expected long-term return from a given state and guides policy improvement. In standard implementations, the value function is approximated by a neural network trained from scratch. This *tabula rasa* approach disregards available domain knowledge, forcing the agent to discover the underlying

dynamics and task structure through extensive and often costly trial-and-error interaction with the environment.

In contrast, classical control theory provides a mature framework for analyzing and controlling dynamical systems, often leveraging analytical models. Lyapunov stability theory, in particular, offers a rigorous method for certifying system stability around an equilibrium point. This is achieved through a Lyapunov function: a scalar, energy-like function of the system's state that is positive definite and decreases along all system trajectories. The existence of such a function provides a formal guarantee of convergence to the equilibrium. For many physical systems, an analytical Lyapunov function can be derived from first principles, representing a potent piece of prior knowledge about the system's behavior.

The objective of a Lyapunov function—to decrease towards a minimum at a stable equilibrium—bears a strong conceptual resemblance to the optimal value function in a reinforcement learning stabilization task. This parallel suggests a promising opportunity to fuse the strengths of both domains: using a known Lyapunov function as a structural prior to guide the value function learning process. Instead of requiring a neural network to learn the entire value landscape from a random initialization, we can initialize it with a physically meaningful approximation. The learning problem is thereby reframed from discovering the full function to learning a smaller, potentially simpler residual correction that accounts for model inaccuracies or complex dynamics not captured by the analytical form.

In this work, we investigate this hypothesis by structuring the critic's value function as the sum of a known analytic Lyapunov function and a learned neural network residual. We formalize this decomposition as $V(s) = \Phi(s) + f_\theta(s)$, where $\Phi(s)$ is the pre-defined Lyapunov function and $f_\theta(s)$ is the output of the neural network. To align the reinforcement learning objective with the control-theoretic goal of stabilization, we define the reward signal as the decrease in the Lyapunov function between steps, $R_t = \Phi(s_t) - \Phi(s_{t+1})$. We implement this approach using the Proximal Policy Optimization (PPO) algorithm on the classic pendulum swing-up and stabilization task. By comparing an agent with this Lyapunov-structured critic against a standard PPO baseline, we aim to quantify the impact of this structural prior on the convergence speed of the value function approximation and to assess whether accelerated critic learning translates into improved final policy performance.

2 Methods

2.1 Environment and task formulation

All experiments were conducted using the `Pendulum-v1` environment from the Gymnasium library. The task is to swing up an underactuated pendulum and stabilize it in the upright position. The state is represented by $s = [\cos\theta, \sin\theta, \dot{\theta}]$, where θ is the angle from the vertical upright position and $\dot{\theta}$ is the angular

velocity. The action is a continuous torque applied to the pendulum's joint.

To align the reinforcement learning objective with the control-theoretic goal of stabilization, we replaced the environment's default reward function with a custom reward signal derived from a Lyapunov function. The Lyapunov function, representing the mechanical energy of the pendulum relative to the upright equilibrium, is defined as:

$$\Phi(s) = (1 - \cos\theta) + 0.5\dot{\theta}^2 \tag{1}$$

The reward at each timestep t is then defined as the decrease in this energy function:

$$R_t = \Phi(s_t) - \Phi(s_{t+1}) \tag{2}$$

This formulation incentivizes the agent to find policies that consistently reduce the system's energy, thereby driving the state towards the stable equilibrium at $(\theta, \dot{\theta}) = (0, 0)$. Episodes were terminated after 200 timesteps.

2.2 Reinforcement learning algorithm

We employed the Proximal Policy Optimization (PPO) algorithm, an on-policy actor-critic method. The actor (policy) and critic (value function) were implemented as separate neural networks, each consisting of two fully-connected hidden layers with 64 units and Tanh activation functions. The actor network outputs the mean and standard deviation of a Gaussian policy.

Training was performed using a learning rate of 3×10^{-4} for both networks. Key PPO hyperparameters were held constant across all experiments: a discount factor $\gamma = 0.99$, a Generalized Advantage Estimation (GAE) parameter $\lambda = 0.95$, a clipping ratio $\epsilon = 0.2$, and an entropy coefficient of 0.01. Data was collected in rollouts of 2048 steps, and for each rollout, the networks were updated for 4 epochs using minibatches of size 64.

2.3 Value function architectures

To investigate the impact of incorporating a control-theoretic prior, we compared two distinct critic architectures. The actor architecture was identical in both conditions.

2.3.1 Baseline critic

The baseline condition (Condition A) utilized a standard PPO critic. The neural network was trained to directly approximate the state-value function $V(s)$ by minimizing the error between its predictions and the empirical returns computed from environment interaction.

2.3.2 Lyapunov-structured critic

The experimental condition (Condition B) employed a critic with a structured value function. The value function was decomposed into a known analytical

component and a learned residual component:

$$V(s) = \Phi(s) + f_\theta(s) \tag{3}$$

Here, $\Phi(s)$ is the fixed, non-trainable analytical Lyapunov function defined in Equation (1), and $f_\theta(s)$ is the output of a neural network with the same architecture as the baseline critic. The network is trained only to learn the residual correction $f_\theta(s)$, effectively using the Lyapunov function as a physically-informed baseline for the value estimate.

2.4 Training and evaluation

Both conditions were trained for a total of 100,000 environment steps. To account for stochasticity, we conducted 5 independent training runs for each condition, each with a different random seed.

We evaluated performance using several metrics:

- **Learning Curves:** We tracked the mean episode return (the sum of Lyapunov-based rewards over an episode) as a function of training steps, smoothed over a rolling window of 10 episodes.
- **Critic Loss:** We measured the mean squared error between the critic's value predictions and the GAE-computed returns at each training update. This metric directly quantifies the critic's learning progress.
- **Value Function Analysis:** After training, we evaluated the learned value functions from a representative seed on a 100×100 grid of states spanning $\theta \in [-\pi, \pi]$ and $\dot{\theta} \in [-8, 8]$. We computed the Mean Squared Error (MSE) between each learned value function and the analytical Lyapunov function $\Phi(s)$ over this grid.
- **Upright Stability:** To assess final policy quality, we performed 10 evaluation rollouts per seed using the deterministic policy (i.e., taking the mean of the policy's action distribution). We measured stability as the fraction of timesteps in these rollouts where the pendulum was near the upright position. This was defined by $|\theta| < 0.1$ radians, where the angle θ was recovered from the state vector using the `atan2` function.

3 Results

3.1 Policy performance and sample efficiency

To evaluate the impact of the Lyapunov-structured critic on policy learning, we first examined the mean episode return throughout training. As shown in Figure 1, the learning curves for the standard critic (Condition A) and the structured critic (Condition B) were nearly indistinguishable over the 100,000-step training horizon. Both conditions exhibited a gradual increase in returns

during the first 50,000 steps before plateauing at a similar negative mean episode return of approximately -0.78. The variance across the five random seeds was also comparable for both conditions, indicating that the structural prior did not reduce the sensitivity of policy learning to random initialization.

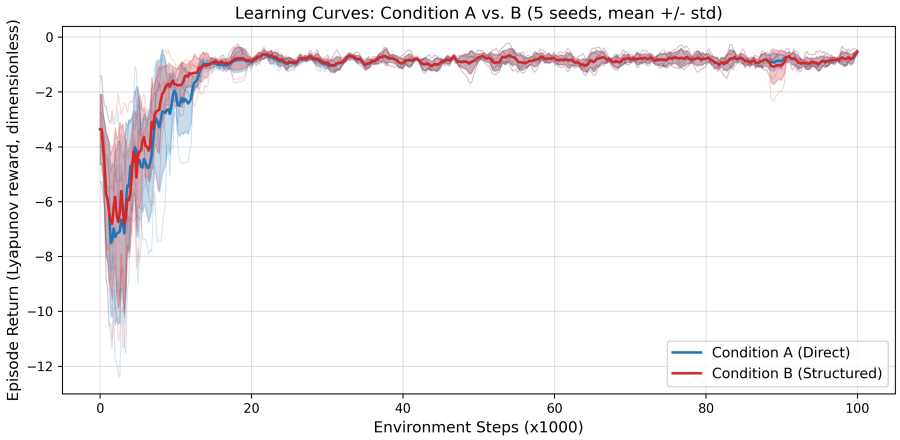

Figure 1: Smoothed episode return versus environment steps for a standard PPO critic (Condition A) and a critic structured with a Lyapunov prior (Condition B). Curves show the mean $\pm$ one standard deviation over five random seeds, with individual seed trajectories overlaid. Both conditions exhibit nearly identical learning trajectories, improving initially before plateauing at a similar negative return value. This suggests that the structural prior in the value function did not lead to a discernible improvement in policy learning speed or final performance within the 100,000-step training horizon.

Consistent with the learning curves, we found no difference in sample efficiency. We defined sample efficiency as the number of steps required to reach 90% of the maximum smoothed return achieved by a given seed. For all five seeds in both conditions, this value was 100,000 steps, the full training budget. This indicates that neither agent's policy had fully converged within the allotted training time.

Finally, we assessed the quality of the final policies by measuring their ability to stabilize the pendulum in the upright position ($|\theta| < 0.1$ radians) during 10 evaluation rollouts. Both agents performed poorly on this metric. The standard agent (Condition A) maintained an upright position for a mean fraction of 0.0010 ± 0.0018 of the timesteps, while the structured agent (Condition B) achieved a fraction of 0.0041 ± 0.0038. Although the structured agent's stability was numerically higher, the absolute values are extremely low and the standard deviations are large relative to the means, making the difference practically insignificant. Neither policy learned to reliably stabilize the pendulum.

3.2 Value function approximation

While the structural prior did not translate to improved policy performance within the training horizon, it had a dramatic effect on the critic's learning process and the quality of the final value function approximation. The most direct measure of critic learning is the mean squared error between its value predictions and the GAE-computed returns, which is the loss function the critic minimizes.

Table 1: Comparison of Critic Training Loss (MSE vs. GAE Returns) between the standard (A) and structured (B) critics.

Training Phase	Condition A	Condition B
Early (first 20%)	5.686	**0.733**
Overall Mean	1.057	**0.136**
Final (last 20%)	**0.0010**	0.0029

As summarized in Table 1, the Lyapunov-structured critic (Condition B) achieved a substantially lower training loss throughout the learning process. The overall mean loss for Condition B was 0.136, an 87% reduction compared to the 1.057 loss of the standard critic (Condition A). The benefit was most pronounced in the early stages of training (the first 20% of updates), where the structured critic's loss was over 8 times lower than the baseline (0.733 vs. 5.686). This demonstrates that the Lyapunov function provides a much better initial estimate of the true value function than a randomly initialized network, significantly accelerating the initial convergence of the critic. By the end of training, both critics converged to very low loss values, with the baseline critic achieving a slightly lower final loss.

To further analyze the learned value functions, we evaluated them on a 100×100 grid of states after training. We compared the learned value functions, $V_A(s)$ and $V_B(s)$, to the analytical Lyapunov function $\Phi(s)$. The Mean Squared Error (MSE) between the standard critic's value function and the Lyapunov function was $\text{MSE}(V_A, \Phi) = 71.98$. In contrast, the structured critic's value function had an MSE of only 10.07, an 86% reduction. This confirms that the structural prior successfully constrained the learned value function to be much closer to the analytical, energy-based form.

It is important to note that the residual network $f_\theta(s)$ in the structured critic did not converge to zero. The MSE of 10.07 for Condition B is entirely attributable to this learned residual. We found that the mean absolute value of the residual, $|f_\theta(s)|$, across the state grid was approximately 26% of the mean value of the Lyapunov function $\Phi(s)$. This indicates that the network learned a non-trivial correction to the Lyapunov prior, likely to account for the value function of the agent's suboptimal policy, which deviates from the ideal energy landscape.

4 Conclusions

This paper investigated the challenge of sample inefficiency in deep reinforcement learning for continuous control, focusing on the difficulty of learning accurate value functions from scratch. We proposed and evaluated a method to accelerate critic learning by incorporating control-theoretic domain knowledge. This was achieved by structuring the critic's value function as the sum of a known, analytical Lyapunov function and a learned neural network residual.

Our methodology involved applying the Proximal Policy Optimization (PPO) algorithm to the Gymnasium Pendulum-v1 stabilization task. To align the learning objective with the control goal, we defined the reward signal as the decrease in the system's energy, as described by the Lyapunov function. We then compared the performance of a standard PPO agent against an agent equipped with our proposed Lyapunov-structured critic over a 100,000-step training horizon.

The results demonstrated a clear and significant benefit of the structural prior on the critic's learning process. The Lyapunov-structured critic achieved an 87% lower overall training loss and converged substantially faster, particularly in the early stages of training where its loss was 8 times lower than the baseline. Furthermore, the final learned value function was 86% closer to the analytical Lyapunov function, confirming that the prior effectively guided the approximation. However, these dramatic improvements in value function approximation did not translate into superior policy performance or sample efficiency. The learning curves and final policy stability metrics were nearly identical for both agents, and neither learned to reliably stabilize the pendulum within the given training budget.

From these findings, we conclude that incorporating a Lyapunov function as a structural prior is a highly effective technique for accelerating the convergence of the value function in an actor-critic framework. It provides a strong, physically-meaningful initial estimate that the neural network only needs to refine. However, we also learned that for on-policy algorithms like PPO, rapid critic convergence does not guarantee a corresponding acceleration in policy improvement, at least not on shorter training horizons. The policy update mechanism may represent a separate bottleneck that requires more extensive environment interaction to leverage the more accurate value estimates provided by the structured critic. This suggests a potential decoupling between the learning speeds of the actor and the critic, a factor that warrants further investigation in the pursuit of more sample-efficient reinforcement learning.